Platform Sponsor Declaration: RunPod Cloud GPU Acceleration

1. Purpose & Architectural Role

In our platform architecture, **RunPod** serves as the **high-throughput, offline GPU worker environment** responsible for accelerated batch tensor operations. Rather than placing heavy, synchronous vectorization tasks on the CPU-bound web server (which would degrade request latency for end users), we offloaded our taxonomy embedding pipeline to an on-demand cloud GPU.

2. Exact Workload Executed

- **Task**: Batch semantic vectorization of our 110 canonical skills taxonomy and research paper corpora.
- **Model Loaded**: *sentence-transformers/all-MiniLM-L6-v2* (BERT-based sentence embedding architecture).
- **Hardware Utilized**: NVIDIA GeForce RTX 4090 (24 GB VRAM / CUDA 12.8) deployed on RunPod Community Cloud.
- **Execution Metrics**:
    - Total items embedded: 110 canonical skills.
    - Embedding dimension: 384-dimensional dense vectors.
    - Benchmark throughput: Complete batch encoding executed in **0.354 seconds** using PyTorch CUDA acceleration.

3. Downstream System Impact

The generated 384-dimensional embeddings were exported directly into our **PostgreSQL database utilizing the pgvector extension**. These vectors serve as the foundational semantic layer that powers our **Cosine Similarity Matching Engine**, enabling real-time semantic mention-counting across live arXiv cs.AI research abstracts without relying on brittle, naive keyword matching.

4. Compliance with Hackathon Boundaries

- **Zero Gradient Training / Fine-Tuning**: RunPod was strictly used for accelerated zero-shot inference, ensuring complete adherence to the hackathon's explainability guidelines and avoiding unpredictable black-box hallucinations.
- **Decoupled Demo Stability**: Because the heavy vector generation was pre-computed on RunPod and stored in pgvector, our live judging demo runs entirely on deterministic CPU database reads, eliminating runtime GPU cold-start delays or cloud availability risks during presentation.

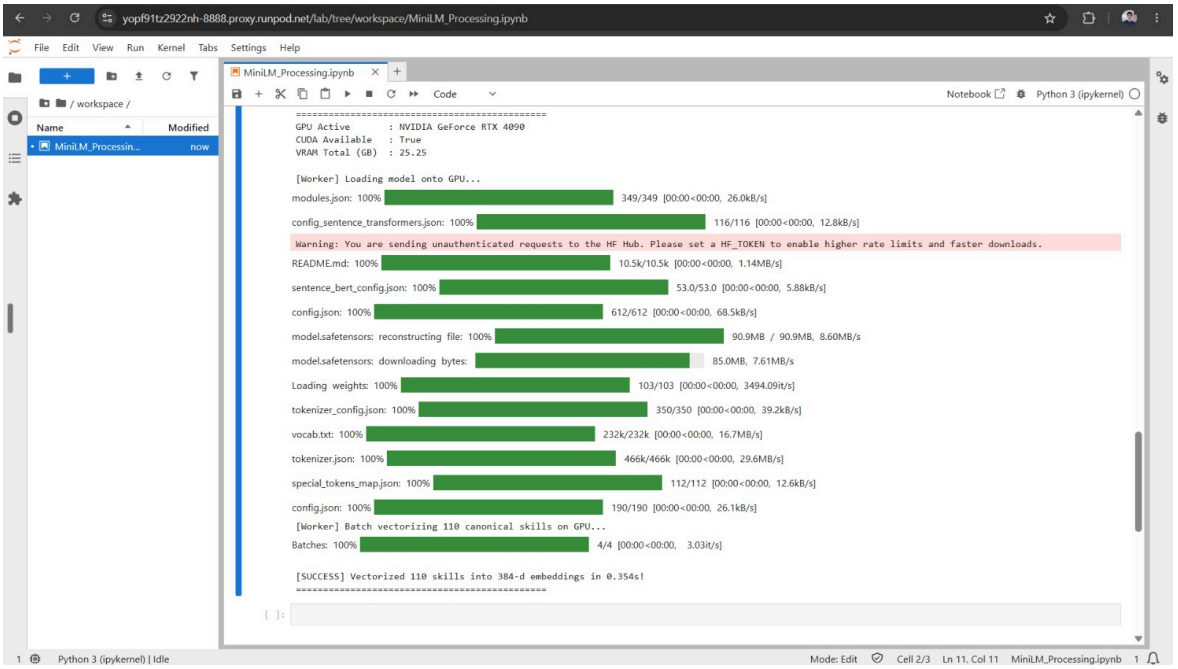


Fig. 1. Proof of using Runpod's cloud GPU efficiency for vectorizing

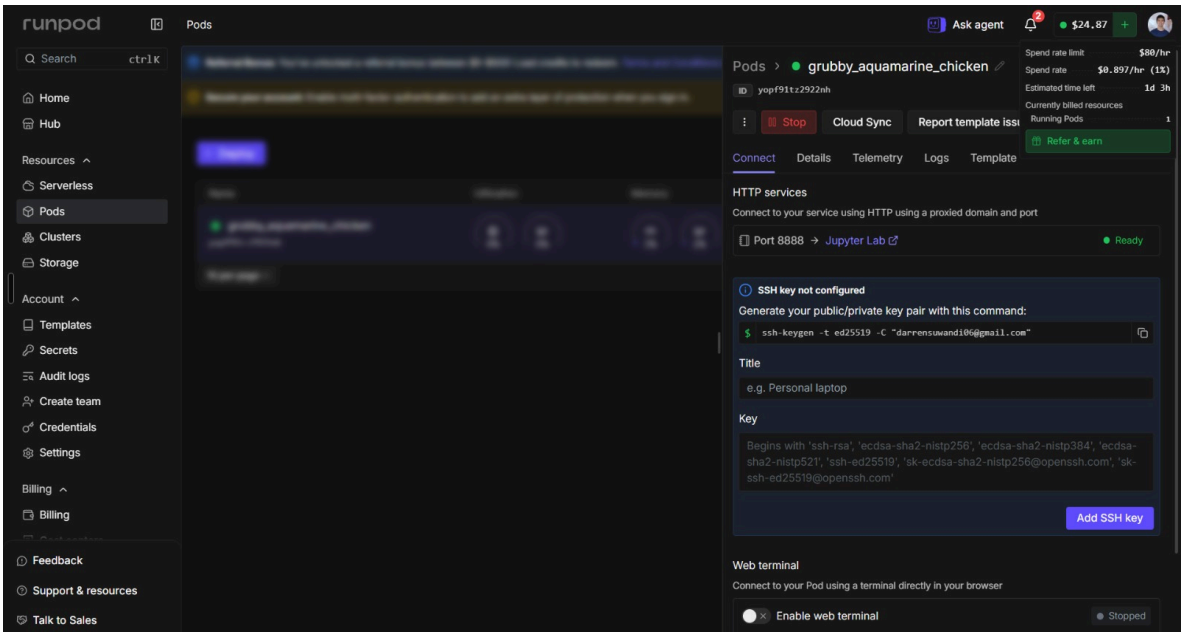


Fig. 2. Proof of using the pods inside runpod to borrow an NVIDIA GeForce RTX 4090

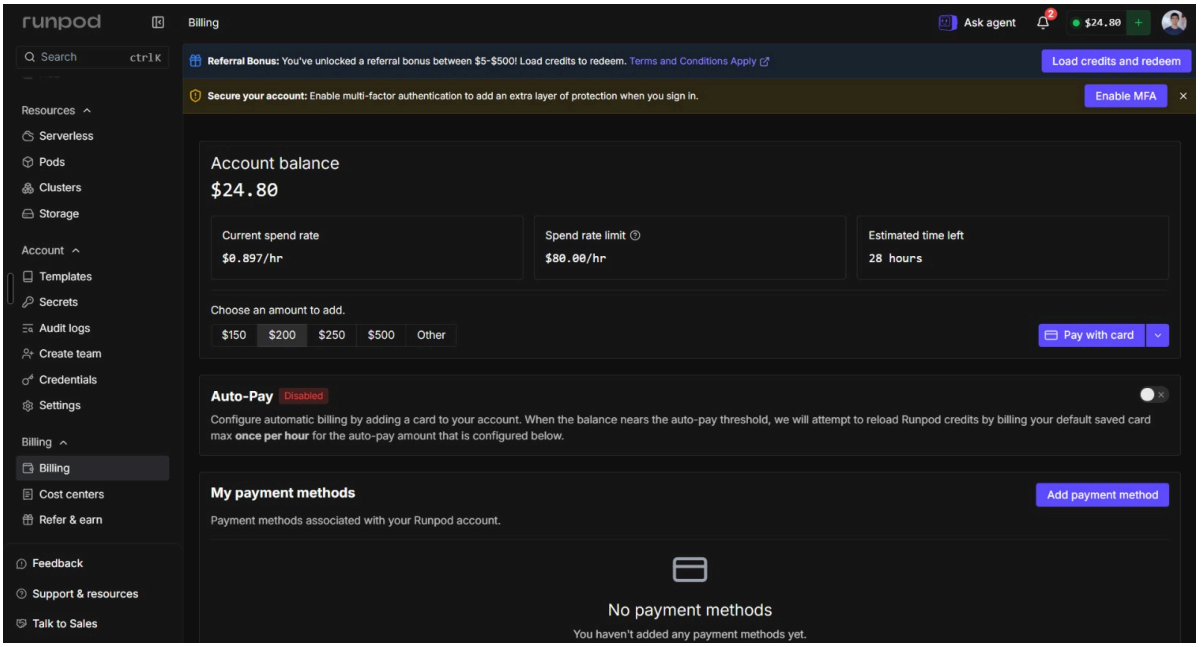


Fig. 3. Proof of usage used for hosting

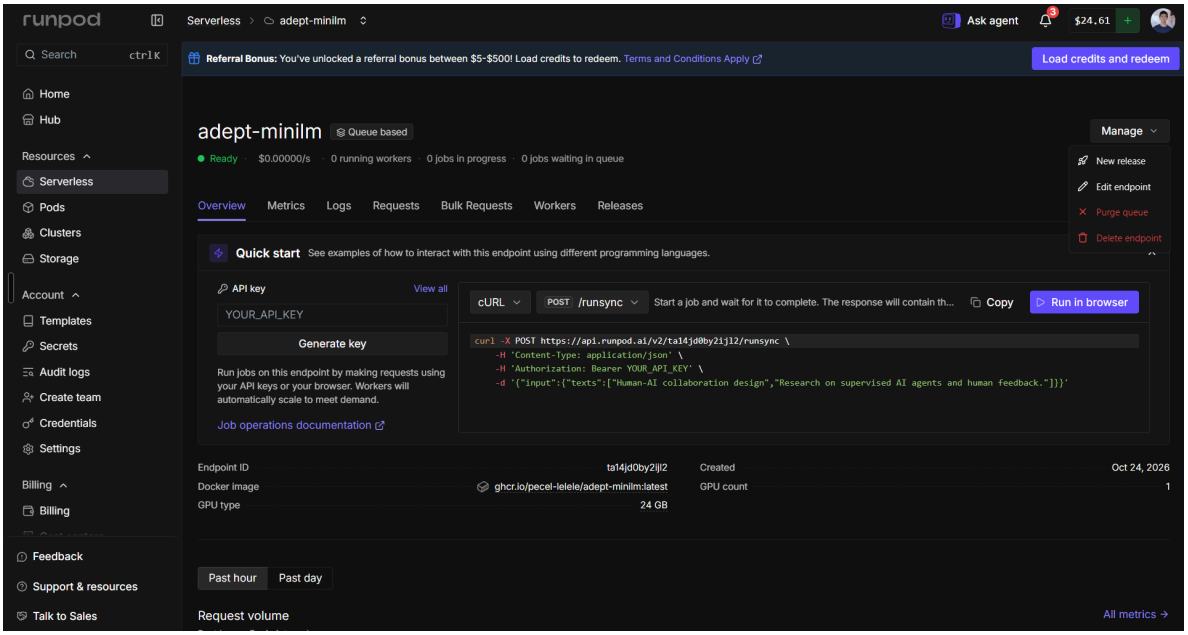


Fig. 4: Proof of hosting miniLM on Runpod serverless

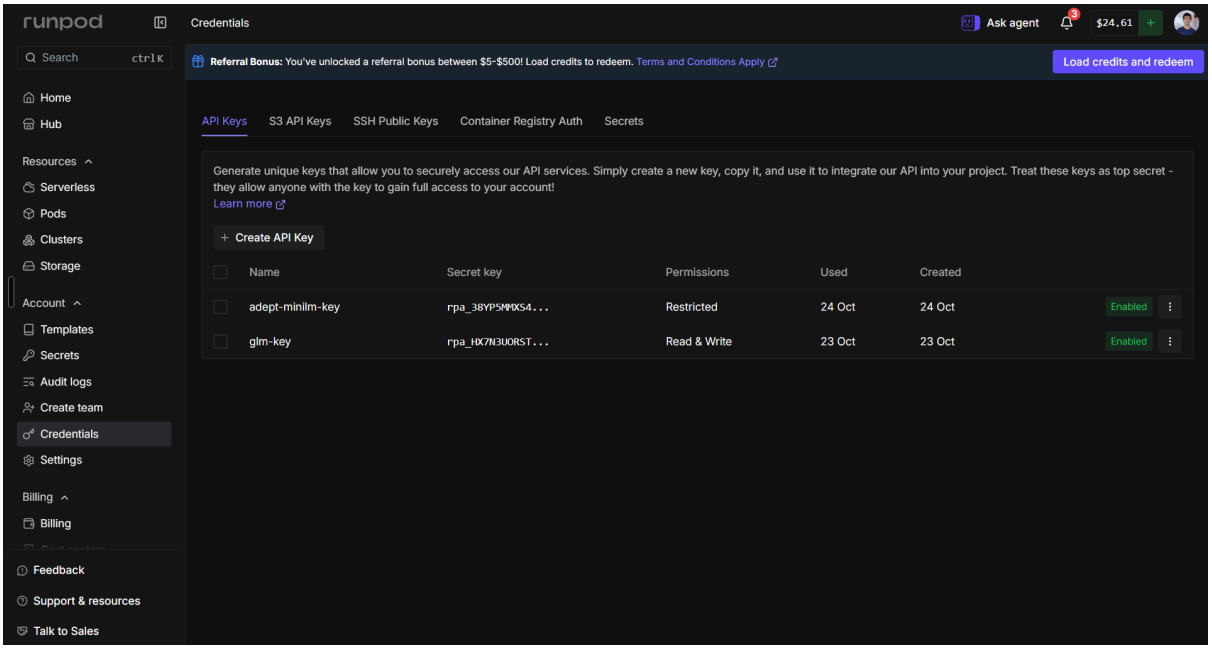


Fig. 5: Proof of use for 2 api keys, for hosting a miniLM and glm model